# Gradient Descent: Theory and Applications

*A comprehensive guide with formulas, tables, and illustrations*

Detailed document covering types of gradient descent, mathematical foundations, and visual explanations.

Autor: Maru

February 2026

## Introduction to Gradient Descent

Gradient descent is an optimization algorithm used to minimize a function by iteratively moving towards the steepest descent direction as defined by the negative of the gradient. It is widely used in machine learning and deep learning to minimize cost functions and improve model accuracy.

## Mathematics Behind Gradient Descent

The core idea of gradient descent is to update parameters \( \theta \) in the direction that reduces the cost function \( J(\theta) \). The update rule is given by:

\[ \theta := \theta - \alpha \nabla J(\theta) \]

where:
- \( \theta \) represents the parameters to optimize.
- \( \alpha \) is the learning rate, controlling the step size.
- \( \nabla J(\theta) \) is the gradient of the cost function with respect to \( \theta \).

$$\theta := \theta - \alpha \nabla J(\theta)$$

Gradient Descent Update Rule

This iterative process continues until convergence, ideally reaching a minimum of the cost function. The gradient points in the direction of the steepest ascent, so moving in the opposite direction minimizes the function.

## Types of Gradient Descent

- Batch Gradient Descent: Uses the entire dataset to compute the gradient at each step. It is stable but can be slow for large datasets.
- Stochastic Gradient Descent (SGD): Uses one data point at a time to update parameters. It is faster but introduces noise in the updates.
- Mini-batch Gradient Descent: Uses a small random subset (mini-batch) of the data to compute the gradient, balancing speed and stability.

Comparison of Gradient Descent Types

| Type | Description | Advantages | Disadvantages |
|---|---|---|---|
| **Batch Gradient Descent** | Uses all data for each update | Stable convergence | Slow for large datasets |
| **Stochastic Gradient Descent** | Updates per single data point | Fast and can escape local minima | Noisy updates, less stable |
| **Mini-batch Gradient Descent** | Updates per small data batch | Balances speed and stability | Requires tuning batch size |

## Challenges in Gradient Descent

Gradient descent can face challenges such as local minima, plateaus, and saddle points which can slow down or trap the optimization process. Choosing an appropriate learning rate is crucial to

ensure convergence without overshooting.

## Visual Illustrations

**[Image Placeholder: Gradient descent path on a cost function curve.]**

**Gradient descent path on a cost function curve.**

**[Image Placeholder: Comparison of stochastic, mini-batch, and batch gradient descent.]**

**Comparison of stochastic, mini-batch, and batch gradient descent.**

**[Image Placeholder: Learning steps showing convergence to minimum.]**

**Learning steps showing convergence to minimum.**

**[Image Placeholder: Local minimum and plateau challenges in gradient descent.]**

**Local minimum and plateau challenges in gradient descent.**

**[Image Placeholder: Contour plot showing gradient descent trajectory.]**

**Contour plot showing gradient descent trajectory.**

**[Image Placeholder: Vector representation of the gradient direction.]**

**Vector representation of the gradient direction.**

**[Image Placeholder: Gradient descent path in a contour plot.]**

**Gradient descent path in a contour plot.**